# AAMIR TALAL

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## EDUCATION

**CY Tech, Paris** *2026 – 2029*
Engineering Cycle (M.Eng), Computer Science, **Machine Learning specialisation**
Top 5% of the cohort; machine learning, statistics, optimisation, data science

**CY Tech, Paris** *2024 – 2026*
Integrated Preparatory Cycle: mathematics, physics and computer science

## EXPERIENCE

**Safran Aircraft Engines** *2026 – 2029*
*AI Engineer, Applied Research (Apprenticeship)* *Montereau-sur-le-Jard*

- Design and deploy LLM agents that automate and digitalise sales administration (ADV) processes
- Build and evaluate agentic workflows that ground generative AI in existing business tools and data
- Work directly with business teams to identify, scope and prioritise automation use cases

## APPLIED RESEARCH PROJECTS

**Remaining Useful Life Prediction with Calibrated Uncertainty** *2026*
*Predictive maintenance on NASA C-MAPSS turbofan engines*

- Reproducible RUL study on all four C-MAPSS subsets (FD001–FD004: up to 6 operating conditions, 2 fault modes), piecewise-linear target capped at 125 cycles, 30-cycle sensor windows
- Leakage-free protocol: regime-wise sensor normalisation (k-means on operating settings), statistics fitted on training engines only, held-out engines for early stopping, official test set used once
- Benchmarked Ridge and gradient boosting against a 5-member 1D-CNN deep ensemble trained with Gaussian NLL, decomposing aleatoric and epistemic uncertainty; scored with RMSE and the asymmetric NASA score
- Gradient boosting best at 12.4–14.1 test RMSE; the ensemble's 90% intervals cover 88–93% of test engines; CNN reaches 14.6 RMSE on FD002 vs. 22.4 reported for the same backbone (Li et al., 2018)
- Stack: Python, PyTorch, scikit-learn, pandas

**Legible: speaker-attributed grounding for legal LLMs** *2026*
*Winner, Mistral×Stanford Law School Hackathon*

- Designed the System 1→System 2 engine: rule-based zoning and a calibrated classifier (Jev) attribute each sentence of a French court ruling to its speaker with a confidence score, conditioning a Mistral LLM that answers with paragraph-level citations
- On the team's 25-ruling benchmark, 4× fewer misattributed answers than a baseline LLM (11 vs 44 out of 224)
- Stack: Python, FastAPI, Mistral API, MCP; live demo: legible-or4n.onrender.com

**MiroPolis: multi-agent LLM simulation of parliament** *2026*
*Top Project, National Assembly Hackathon*

- Built a multi-agent digital twin (OASIS) in which LLM deputies, seeded with the chamber's real composition, groups and programmes, debate, amend and vote on a bill
- Compared fine-tuned agents with persona-orchestrated pre-trained LLMs, and pivoted to the latter as faster and more realistic

## TECHNICAL SKILLS

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|---|---|
| **Machine Learning** | Deep ensembles, uncertainty estimation, 1D-CNNs, gradient boosting, PyTorch |
| **AI / LLM** | LLM agents, RAG, prompt engineering, LLM evaluation, MCP, LangChain, Hugging Face |
| **Data** | SQL, pandas, NumPy, scikit-learn, data pipelines, n8n |
| **Languages** | Python, C, SQL, JavaScript, Shell/Bash, PHP |
| **Spoken** | French (native), English (C1), Urdu (C1) |

## HACKATHONS & ACHIEVEMENTS

- Winner, Mistral×Stanford Law School Hackathon, and winner of the Legora track
- Top Score, IBM×Airbus Hackathon (Télécom Paris)
- Top Project, French National Assembly Hackathon
- Top 20 out of 300, Anthropic×Hugging Face Hackathon
- ML algorithms and maths implemented from scratch on Deep-ML; 50+ LeetCode problems solved

## COMMUNITY & INTERESTS

- Content creator, Intellectuel Athlète (personal development / tech): 14,000+ followers, 2–3M monthly views
- Volunteering: 50+ weekly humanitarian outreach rounds (maraudes), Au cœur de la précarité; created an Arabic-learning web app (1,000+ users in 24 h)
- Sport: Brazilian jiu-jitsu and grappling (competition), running